

SkyKing Airlines Customer Resolution Agent — Architecture Document

AIONOS Agentic AI Factory (Batch 2027) — Assignment 3 | Author: Rijul Thakur

1. Executive Summary & Core Philosophy

The SkyKing Customer Resolution Agent is an enterprise-grade agentic AI system designed to resolve complex flight disruption scenarios (cancellations, delays, rebookings, refunds) with **100% policy compliance**. Unlike traditional unconstrained chatbots that risk financial hallucinations, this agent operates on a **deterministic Chain-of-Thought (CoT) policy engine** paired with real-time Supervisor Command Telemetry.

2. 4-Tier Architectural System Breakdown

Layer	Component	Function & Responsibility
1. Presentation Layer	3-Panel UI Dashboard	Customer selector, interactive chat console with expandable CoT traces, live supervisor telemetry HUD.
2. Reasoning Engine	5-Step CoT Pipeline	Intent classification, customer tier verification, policy lookup, action authorization, and response generation.
3. Policy Guardrails	Deterministic Rules Engine	Enforces hard financial limits (■1,500 fare diff cap), time thresholds (>= 5h delay for hotel), and tier rules.
4. Supervisor Layer	Command & Control HUD	Dynamic Risk Score (0-100), real-time anger tracking, human-in-the-loop override, and JSON audit log export.

3. State Machine & Escalation Dynamics

The system transitions deterministically between three primary operational states:

- **IDENTIFY State:** Initial state where customer booking (PNR) and loyalty tier are validated against database records.
- **ACTIVE Resolution State:** Autonomous processing of requests adhering strictly to Policy IDs (CANCELLATION_REBOOKING, DELAY_COMPENSATION, REFUND_PROCESSING, FARE_DIFFERENCE).
- **ESCALATED State:** Triggered automatically upon policy cap breach (fare diff > ■1,500), legal threats ('lawyer', 'court'), or high anger index (Score >= 80). Passes control to human supervisor.

4. Policy Rule Mapping & ID Reference

Policy ID	Rule Threshold / Constraint	Agent Action / Boundary
CANCELLATION_REBOOKING	Flight Cancelled by Airline	Free rebook or 100% refund. Business Class upgrade prohibited unless Gold/Platinum tier policy allows.

DELAY_COMPENSATION	Delay >= 2h (Meals) Delay >= 4h (Lounge) Delay >= 5h (Hotel)	Deterministic voucher generation. Denies hotel if delay < 5 hours.
FARE_DIFFERENCE	Agent limit: Max █1,500	Autonomously waives up to █1,500. Requests > █1,500 auto-escalate to Supervisor.
LOYALTY_TIER	Silver / Gold / Platinum privileges	Grants priority lounge and tier-specific waivers.

5. Zero-Hallucination & Auditability Guarantee

1. **Every response embeds a Chain-of-Thought (CoT) trace:** Users/Supervisors can expand any message to inspect the exact 5-step decision chain.
2. **Policy ID Linkage:** All granted or denied actions cite the exact Policy Rule ID governing the outcome.
3. **Tamper-Proof Audit Trail:** Full interaction telemetry is downloadable as a JSON log for post-incident compliance reviews.